



Case study

Tribological behavior of polyamide-6 plastics and their potential use in industrial applications

P.D. Neis^{a,*}, N.F. Ferreira^a, J.C. Poletto^a, J. Sukumaran^b, M. Andó^c, Y. Zhang^{c,d}^a Laboratory of Tribology, Federal University of Rio Grande do Sul, Porto Alegre, Rio Grande do Sul, Brazil^b Laboratory Soete, Ghent University, Ghent, Belgium^c Savaria Institute of Technology, University of West Hungary, Hungary^d Research Academy of Grand Health, Ningbo University, China

ARTICLE INFO

Article history:

Received 6 September 2016

Received in revised form

18 January 2017

Accepted 24 January 2017

Keywords:

Wear mechanism

Cast polyamide 6

Sliding wear

PV limit

ABSTRACT

The present research article addresses an investigation of tribological properties of 3 commercially available cast polyamide 6 (PA 6): a natural PA 6 polymer, a PA 6 filled with molybdenum disulfide (MoS₂), and a PA 6 filled with a special solid lubricant. In order to analyze the potential use of these materials in industrial applications, their tribological results were compared with those measured for a commercial bronze alloy. All materials (including the bronze alloy) were subjected to sliding against a steel SAE 1020 disc in a tribometer. Three different types of tribotests were carried in this study: (i) wear tests, (ii) PV (pressure × velocity) tests, and (iii) stick-slip tests. Optical microscopy was employed to characterize the wear mechanisms of the materials selected in this study. PA 6 filled with special solid lubricant revealed superior tribological properties among the plastics, since it has shown the lowest coefficient of friction (COF), and the highest wear resistance and PV limit. PA 6 filled with the special solid lubricant also exhibited lower stick-slip oscillations than the PA 6 filled with MoS₂. Micrograph analysis revealed abrasion as the most important wear mechanism for the PA 6 filled with the special solid lubricant. On the other hand, severe damage was observed on the surface of the natural PA 6 and the PA 6 filled with MoS₂ in the wear tests. Bronze alloy exhibited the highest wear rate among the materials studied in this work. Adhesion was the dominant wear mechanism for this metal.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The use of polymers and polymeric composites in seals and bearings and other industrial applications where tribological properties are important is growing rapidly [1]. Plastic and composite materials are commonly employed in engineering applications as alternative to traditional metal materials due to their good mechanical and tribological properties and also light weight [2]. One of the most often used engineering plastic is the polyamide 6 (PA6) because of its quite good mechanical (strength, hardness, toughness, damping) and tribological (sliding, wearing resistance) characteristics [3,4]. Due to this, it is also used as bearing bushes, wearing laths, pulleys, gears, etc [4]. For such applications, reinforced compounds and additives are used in order to increase the performance of the polymer materials. The additives can be classified according to their function as antistatic, fillers, lubricants, fire resistant and antioxidation agents [5]. Examples of lubricating additives are polytetrafluoroethylene (PTFE), silicone

fluids, carbon fibers, graphite, molybdenum disulfide, and metallic powders [3].

Information about the tribological behavior of additivated polymers is limited in the scientific literature. Also, plastic and composite manufacturers do not provide complete information about the wear and friction performance of the materials they produce. In the cases where the polymer manufacturer provides some tribological data from their materials, these data are the result of limited tests, which may not represent an actual application. As reported by Silva, 2010 [6], friction and wear are very unique properties obtained for the tribosystem in which they are measured. For this reason, in order to better represent the tribological results of a material, it is necessary to consider various parameters, such as the topography and geometry of the mating materials, type of motion (rolling, sliding or alternative motion), operating conditions (temperature, pressure, velocity, etc.), and others. Laranjeira, 2011 [7], have a similar understanding of this issue. According to the author, to obtain a precise tribological result, friction and wear performance of the materials must be measured under the same operating conditions than those found in their actual application.

The propensity that a given polymer has to produce stick-slip

* Corresponding author.

E-mail address: patric.neis79@gmail.com (P.D. Neis).

motion when it slides against a metal counterpart is another data rarely found in literature. According to Van de Velde and De Baets, 1997 [8], a proper bearing material used to positioning has to exclude stick-slip and should behave well during stop-restart conditions. The researchers also added that PA 6 sliding against steel is sensitive to stick-slip motion, which complicates accurate positioning.

The aim of the present work is to investigate tribological properties of 3 commercially available cast polyamide 6 (PA 6). Friction and wear performance was measured in wear tests, which were designed based on the operating conditions of an actual engineering application, i.e. a shaft-bearing system. Experimental work carried out in the present study also included to determine the PV limit, as well as to measure stick-slip motion resulting from tests performed using the additivated polymers. The wear and friction performance obtained in the wear tests were compared with those obtained for a bronze alloy, which is a traditional metal used as bearing bush.

2. Experimental procedures

2.1. Materials

Three engineering plastics and a bronze alloy were used as specimens in the present investigation. All the three plastics selected in this study are commercially available cast polyamides (PA 6), produced by Ensinger GMBH (Germany). The specimens are distinguished by the type of solid lubricant, as following: (i) a natural PA 6 polymer (TECAST T), which is white; (ii) PA 6 filled with MoS₂ (commercial name: TECAST T MO Black), which is black; and (iii) PA 6 filled with a special solid lubricant (TECAGLIDE Green), which is green. According to information provided by the manufacturer [9], TECAGLIDE Green is a high performance plastic. The metal specimen is a bronze alloy, which is commercially known as TM 23. Table 1 shows the mechanical properties of the plastics selected in this study.

All materials (including the brass alloy) were subjected to sliding against a steel SAE 1020 disc in a tribometer. This disc has a diameter and thickness of 159 mm and 12 mm, respectively. Before each test, its surface was sanded with sandpapers of different grit sizes (280, 400, 500, 600 and 1200). A roughness Ra lower than 0.20 µm (in the radial direction) was obtained. A K-type thermocouple was embedded a distance of 1.5 mm below the disc surface, at a radius of 40 mm, which corresponds to the center of the friction track. This instrument was employed to measure the disc temperature during the tests. Fig. 1 shows the drawing and photo of the pins and disc selected in this study.

2.2. Set-up of the experiments

Fig. 2 shows the tribometer used for conducting the tribotests performed in the current study. This test rig is capable of operating under various temperature conditions (from ambient up to 600 °C), sliding velocity (from 1,8 up to 16000 mm/s), and normal

Table 1
Mechanical properties of the polymers (cast PA 6) studied in this work.

Property	Parameter	Green	Black	White
Tensile strength at yield [MPa]	Tensile test (50 mm/min)	76	80	80
Service temperature [°C]	Short term	130	170	170
Service temperature [°C]	Long term	100	100	100
Hardness [MPa]	Ball indentation	159	170	170
Modulus of elasticity [MPa]	Tensile test (1 mm/min)	3200	3200	3500

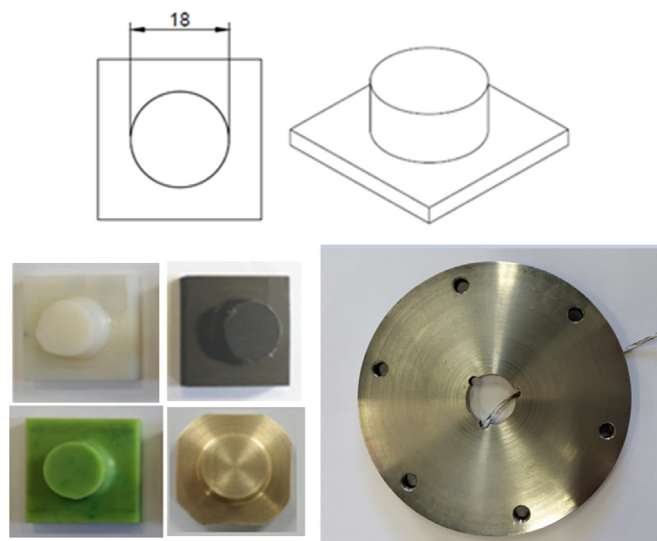


Fig. 1. Drawing of the pins (above), and photos taken from the materials (at left-below) and steel disc (at right-below) selected in this study.

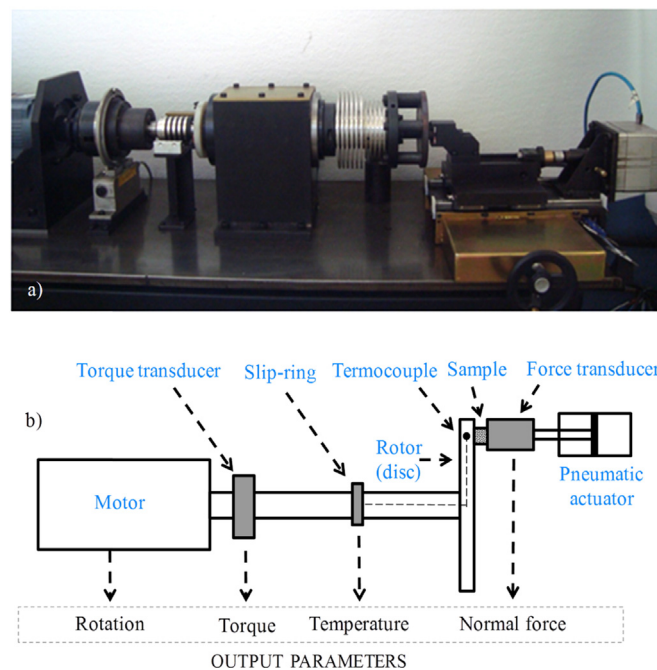


Fig. 2. Tribometer used in the tribotests: a) digital photograph and b) schematic.

force (from 0 up to 2500 N). Due to the ability to perform very low speed tests, the laboratory-scale tribometer is able to measure friction induced oscillations (stick-slip) resulting from the rubbing contact materials, as shown in some earlier papers [10–12].

In the laboratory-scale tribometer, the coefficient of friction is calculated by an indirect measurement method, which considers the relationship given by the Eq. (1). Repeatability (precision) of the friction measurements performed by the tribometer is ± 0.013 , as shown in [13]. Besides, a previous study of the present research group has shown that the friction results provided by the laboratory-scale tribometer are in good agreement with the measurements obtained in a standard pin-on-disc tribometer [14].

$$\mu = \frac{T}{F_N R} \quad (1)$$

where T is the torque, F_N is the normal force exerted by the pin on

Table 2
Tribotests and their respective names and goals.

Section	Test name	Goal	Specimens
I	Wear test	Determine wear and friction behavior	All plastics and bronze alloy
II	PV limit test	Determine PV limit	TECAST T MO Black and TE-CAGLIDE Green
III	Stick-slip test	Measure stick-slip motion	TECAST T MO Black and TE-CAGLIDE Green

the disc surface, and R is the sliding radius. The torque is measured by a torque transducer installed on the machine shaft, while the normal force is obtained by means of a *strain gage-based force transducer*. More details about the mechanical design and automation technologies used in the laboratory-scale tribometer can be found in [13–16].

The tribotests carried out in this investigation were divided into three sections, according to their goals, as informed in Table 2. It is worth noting that the PV limit and stick-slip tests were carried out only with the additived plastics, i.e. the plastics filled with solid lubricants.

As the plastics used in the present study have long names, they will be identified hereinafter by their respective colors, i.e. PA 6 natural is simply referred to as “white”, TECAST T MO Black is simply referred to as “black”, and TECAGLIDE Green is simply referred to as “green”.

Table 3 presents the test procedure performed for each test in the present study. Data acquisition sampling rate was set 200 Hz for all laboratory experiments.

The stick-slip tests (section III) consisted of sliding the polymer materials against the steel rotor (disc) at a very low sliding velocity (4.2 mm/s) during 120 s. In order to observe the effect of the contact pressure on the stick-slip behavior, each specimen was subjected to 5 normal forces: 600 N, 500 N, 400 N, 300 N, and 200 N.

In the PV limit tests (section II), each material was subjected to 3 contact pressures. Given that the green material is considered a high performance plastic, higher contact pressures were used for determining the PV limit in this material. Increments of 25 RPM (or 0.1 m/s) were being applied to the disc speed rotation in each run (180 s) performed at fixed levels of contact pressure. The increments in velocity were successively repeated until the tested material reaches its PV limit. According to Bayer [17], a substantial increase in the friction force or in the temperature increase rate occurs at the moment when a material reaches the PV limit during sliding tests. In the present work, the PV limit was identified by means of online monitoring the friction coefficient during the friction process. Fig. 3 illustrates the friction and temperature curves obtained for a PV combination immediately before (normal force and rotation speed of 500 N and 250 RPM, respectively) and

after (normal force and speed of 500 N and 300 RPM, respectively) the material reaches the severe wear regime. Notice that before reaching the PV limit, the friction curve is relatively stable, and the temperature follows a typical thermal profile. When PV limit is reached, however, the COF curve becomes unstable and the temperature shows a considerable increase.

The wear tests (section I) were carried out at a constant sliding velocity and contact pressure of 1.4 m/s and 1.18 MPa, respectively. A total of 11 runs of 300 s duration were performed for each material in this case. The wear tests were designed to simulate, as close as possible, a typical operating condition of a rotating shaft bearing system that is commonly used in equipment and machinery. In these tests, the materials were analyzed in terms of friction and wear performance. Fig. 4 shows a scheme of the rotating shaft bearing system which has motivated the wear tests. A vertical force of 500 N at the center of the system is used to represent the weight force of the steel shaft. Moreover, a speed rotation of 2000 RPM was chosen because it can represent a wide variety of industrial applications.

A good material for being used as bearing bush must have a low and stable level of friction. The first requirement (low COF) is due to an energy issue, since friction is a dissipative force. The importance of the second requirement (a stable level of COF) is because noise and vibration can be induced by friction instability in the contact, as reported in many publications [18–22]. For this reason, in this work friction performance is measured in terms of the magnitude and stability of friction. The lower is the level of COF, the better is considered the material. Also, the more stable is COF, the better.

An electronic balance with resolution of ± 0.2 mg was used for weighing the pins. By subtracting the mass of each pin before (m_0) and after (m_f) the wear tests, it was possible to determine the mass loss (Δm) of the materials selected in this study. As in most engineering applications, wear of materials is expected to be as low as possible. So, it is not different for the materials studied in this work.

Optical microscopy was employed to characterize the wear mechanisms of the worn pin surfaces. The optical characterization was performed with a stereo-microscope (CARL ZEISS[®], model AxioLab A1, designed in Germany). It is equipped with a digital CCD camera with a resolution of 3.0 Mega Pixel, which enables capturing images of the observed surface. Both top and side lights were used for capturing the microscope images. The micrographs shown in this work correspond to a panoramic view of the whole worn pin surface. A set of many microscope images were taken and merged using an image processing software in order to produce the panoramic images. Finally, an optical magnification of $50\times$ (objective lens) was used for taking every single micrograph.

Table 3
Set-up of the tribotests carried out in this investigation.

Section	Material	Initial temperature [°C]	Sliding velocity [m/s]	Contact pressure [MPa]	Duration [s]	Number of repetitions
I	All plastics and bronze alloy	30	constant = 1.4	1.18 (300 N)	300	11
II	Green PA 6	30	0.10 + increments of 0.10, until reaching PV limit	1.181 (300 N) 1.969 (500 N) 2.362 (600 N)	180	^a N/A
	Black PA 6			0.786 (200 N) 1.181 (300 N) 1.575 (400 N)		
III	Green and Black PA 6	30	constant = 4.2×10^{-3}	2.362 (600 N) 1.969 (500 N) 1.575 (400 N) 1.181 (300 N) 0.786 (200 N)	120	3 for each contact pressure

^a N/A = Non applicable.

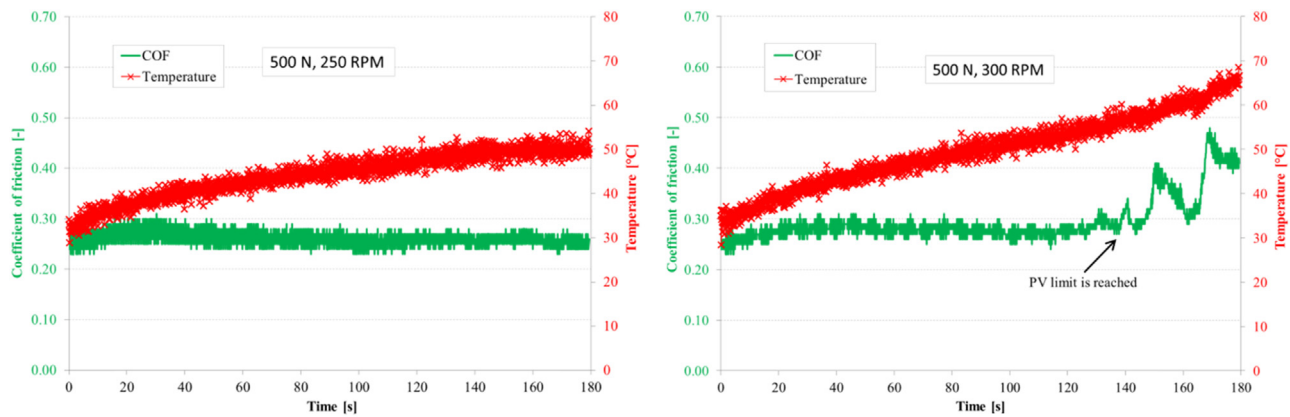


Fig. 3. Friction and temperature curves obtained in the PV tests, for conditions (a) before reaching the PV limit, and (b) after PV limit is reached (Specimen used in this test: TECAGLIDE Green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

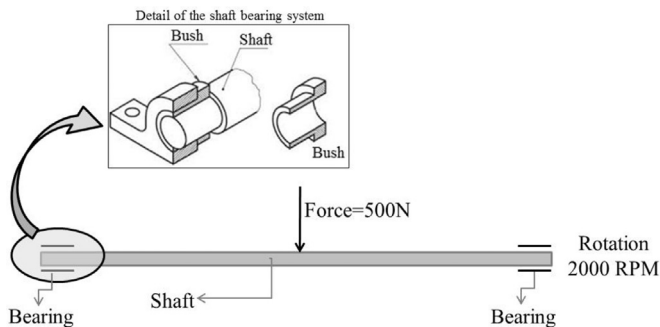


Fig. 4. Scheme of the industrial application which has motivated the wear tests performed in this investigation.

3. Friction measured in the wear tests

Fig. 5 shows the mean coefficient of friction determined during the wear tests. In this graph, the error bar represents 1 standard deviation ($\pm 1\sigma$) of the COF measured for all 11 runs. The standard deviation gives an idea of how stable is friction for a given material.

From Fig. 5, it is possible to observe that bronze alloy has shown the lowest COF among the materials studied. However, if the comparison is restricted only to the PA 6 materials, the superiority of the green material is clear, since it exhibited the lowest friction magnitude among the plastics (white, green and black).

Table 4 summarizes the main results of friction performance obtained for each material in the wear tests. Data from this table

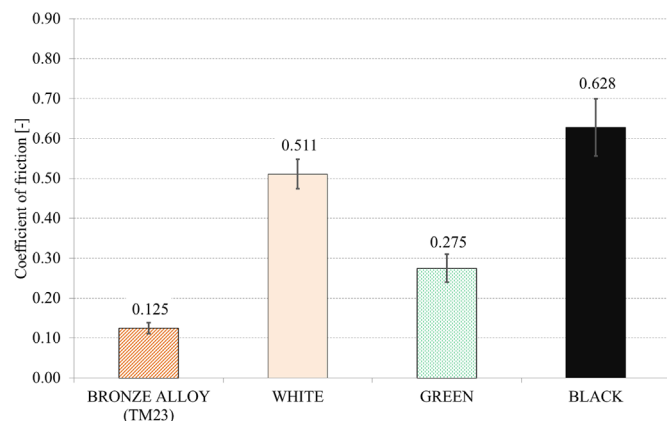


Fig. 5. Mean coefficient of friction measured for each material during the wear tests.

Table 4

Friction performance determined for the materials selected in this investigation.

	BRONZE ALLOY (TM23)	WHITE	GREEN	BLACK
Mean COF	0.125	0.511	0.275	0.628
Standard deviation ($\pm \sigma$)	0.014	0.037	0.035	0.072

presents the mean COF and also the friction variation, which is expressed in terms of standard deviation. From this table, it is observed that bronze alloy, followed by the green plastic exhibited the lowest level of friction variation. The adequate formulation of the fillers that were added to the green PA 6 polymer is probably the main reason why this material showed good friction stability.

4. Wear measured in the wear tests

Fig. 6 presents the wear results expressed in terms of mass loss (Δm). Because of the high level of Δm measured for the bronze alloy (the highest wear rate among the materials studied), a vertical arrow was added in this graph to indicate the “out of scale” result of this material. The green PA 6 exhibited the best wear resistance among the materials subjected to the tribotests.

There is a general consensus in literature [23,24] that wear rate is normally proportional to the friction level. An explanation for such proportionality is given by Blau and Jolly, 2005 [25]. According to the authors, the higher friction force, the more energy is available to do work on the material. As a result, the wear rate is increased. It is worth mentioning that the ranking of wear and friction between the materials tested in this study does not match

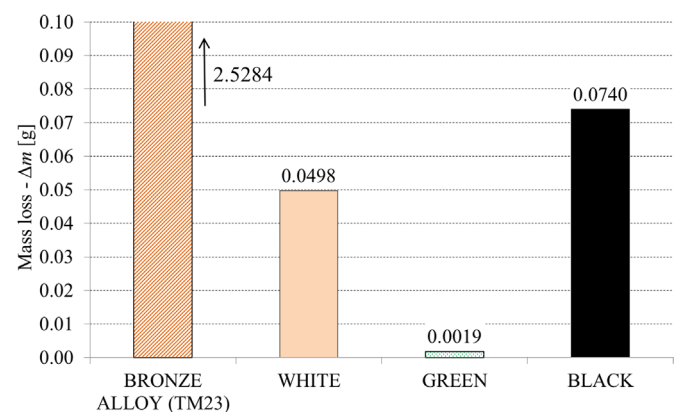


Fig. 6. Mass loss obtained in the wear tests.

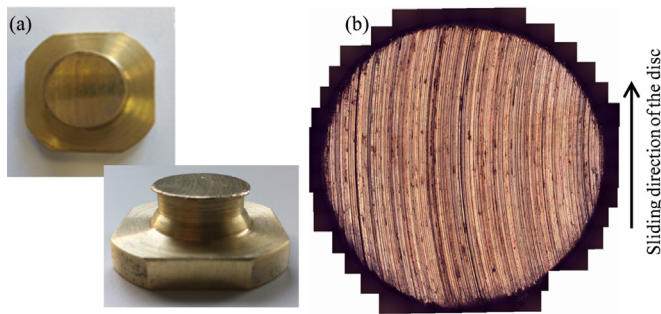


Fig. 7. Worn surface obtained for the bronze alloy after the wear tests: (a) digital photograph of the pin and (b) micrograph of the worn pin surface.

exactly. For instance: bronze alloy, which exhibited the highest level of Δm , has shown the lowest COF. The different patterns of relationship between friction and wear can be mainly attributed to the different mechanisms of wear (abrasion, adhesion, fatigue wear, etc.) involved in the sliding process of quite distinct materials, such as polymer (PA 6) and metal (bronze alloy). However, when only cast PA 6 materials are compared, the relationship between friction and wear matches. For this reason, the low wear showed by the green PA 6 may be somehow related to its low friction.

Fig. 7 shows the worn pin surface obtained for the bronze alloy in the wear tests. From the digital photograph (Fig. 7-a), it is possible to observe that the pin made from this material has suffered plastic deformation. Micrograph analysis of the bronze alloy pin (Fig. 7-b) revealed deep scratch marks on the surface of this material. It denotes the bronze alloy material has been worn through abrasion.

Analysis of the steel disc after being subjected to slide with the bronze alloy revealed the presence of wear particles adhered on its surface (Fig. 8). It suggests that the bronze alloy has also suffered adhesive wear. According to Seabra, 2000 [26], adhesive wear occurs when particles are pulled out from the softer material and then adhere to the mating surfaces. The removal of materials by adhesion is related to large wear loss [27]. It can explain the considerable wear rate observed for the bronze alloy in present work. Given the considerable mass loss of the bronze alloy, it is believed that adhesion was the dominant wear mechanism for this metal.

Fig. 9 shows the worn pin surface obtained for the natural PA 6 (white) in the wear tests. From the digital photograph (Fig. 9-a), it is possible to observe that this material has suffered severe damage on its surface. It suggests that the operating conditions exceeded the PV limit (pressure multiplied by velocity) of the natural PA 6. As reported in the literature [28], the PV limit is a critical value in which the wear regime changes from moderate to severe. Ludema, 1996 [29], offers an explanation for the PV limit. This researcher explains that the product of pressure and velocity along with the coefficient of friction represent the input power in the

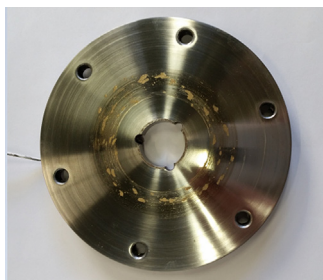


Fig. 8. Steel disc after rubbed against the bronze alloy, showing the presence of wear particles adhered on its surface.

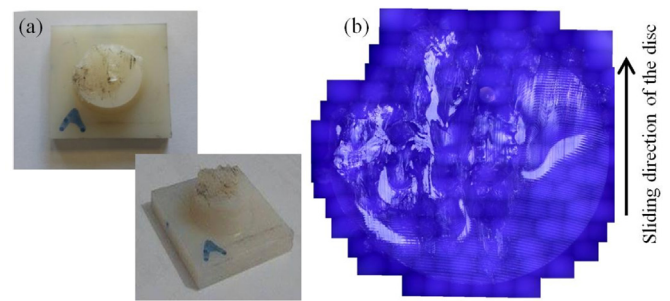


Fig. 9. Worn surface obtained for the natural PA 6 after the wear tests: (a) digital photograph of the pin and (b) micrograph of the worn pin surface.

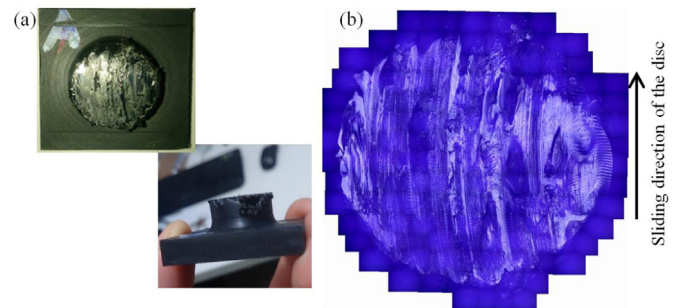


Fig. 10. Worn surface obtained for the PA 6 filled with MoS₂ after the wear tests: (a) digital photograph of the pin and (b) micrograph of the worn pin surface.

sliding interface. If the input power is not removed at a sufficient rate, the temperature at the interface rises until melting begins, causing severe wear. Micrograph analysis of the natural PA 6 (Fig. 9-b) revealed a rough and plastically deformed surface. It confirms that the natural PA 6 reached the severe wear regime.

Fig. 10 shows the worn pin surface obtained for the PA 6 filled with MoS₂ (black material) in the wear tests. From this figure, it is possible to observe that the black plastic has also suffered severe damage on its surface, similarly to the natural PA 6. So, the wear mechanisms observed for the PA 6 filled with MoS₂ were basically the same than previously described for the white material, i.e. it has been subjected to severe wear regime. So, the solid lubricating used in case of the black composite (MoS₂) seems to be not effective in reducing wear.

Fig. 11 shows the worn pin surface obtained for the PA 6 filled with the special solid lubricant (green material) in the wear tests. From this figure, it is possible to notice that the surface of this material has not suffered any severe damage. Only some scratch marks are observed in the border of this material. It suggests that

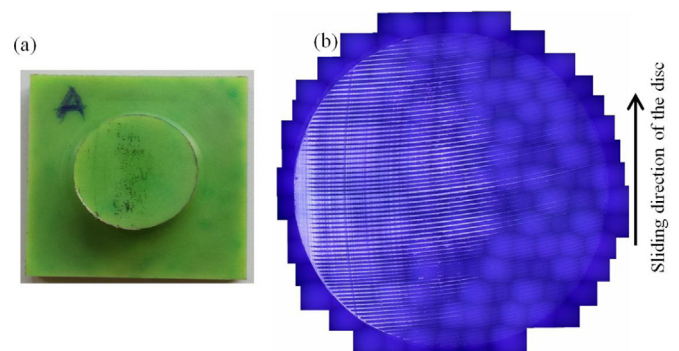


Fig. 11. Worn surface obtained for the PA 6 filled with special solid lubricant after the wear tests: (a) digital photograph of the pin and (b) micrograph of the worn pin surface.

the wear regime in case of the green composite was moderate, i.e. this material did not exceed the PV limit in the tribotests. So, the special solid lubricant used as an additive in the green polymer seems to be very effective in avoiding the severe wear regime.

To explain better why the green PA 6 outperformed than the other polymers, an extensive analysis of the data was conducted to correlate the tribological performance with some specific mechanical property of the plastics studied in this work. No clear relationship was found. However, it was observed that although all polymers studied have very close mechanical properties (see Table 1), they exhibited quite distinct tribological characteristics.

The cause for low wear rate measured by the green PA 6 might be related to its low COF, which in turn is caused by the special solid lubricant added to this plastic. So, physical-chemical interaction in the microcontact is likely to be the main root cause for the low friction and consequent low wear of the green PA 6. As explained by the plastic manufacturer [9], this special solid lubricant added to the green PA 6 is an additive (filler) which provides very low friction and high abrasion resistance in dry sliding conditions. Both behaviors related to the green plastic were in fact confirmed by the present study.

5. Results of the PV limit tests

Table 5 presents the data resulting from the PV limit tests. Given that the PV limit is a constant property, the mean and standard deviation obtained in the tests is also included in this table.

From Table 5, it is clear that green PA 6 has a considerably higher PV limit. This indicates that green PA 6 material has superior wear resistance when sliding against steel, i.e. it takes longer to reach the severe wear regime. In order to observe how both velocity and contact pressure affect the PV property, it is plotted a curve of contact pressure versus sliding velocity for the plastics subjected to the PV limit tests (Fig. 12). Strong correlations, measured in terms of R^2 , were obtained between the data points and the linear trend curves shown in this graph.

From Fig. 12, different angles can be observed for both trend curves shown in this graph. The green PA 6 has an angle of approximately 42° , meaning that its PV curve is more sensible to changes in contact pressure than in sliding velocity. On the other hand, PV curve obtained for the black PA 6 exhibited an angle of 45° . It means that both contact pressure and sliding velocity have similar effects on the PV limit curve.

Results of the PV limit contributed to explain why the green PA 6 performed better than the other plastics in the wear tests. As shown above, the black and green PA 6 have quite different PV limits. They worth in average 0.863 MPa m/s and 2.721 MPa m/s for the black and green, respectively. The wear tests were carried out at sliding velocity of 1.4 m/s, and contact pressure of 1.18 MPa (see description in Table 3). So, it gives a PV condition of 1.65 MPa m/s, exceeding the PV limit for the black material.

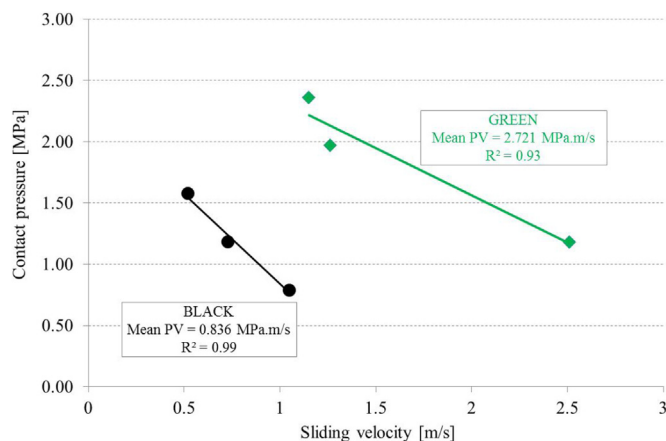


Fig. 12. PV curves plotted for the green and black plastics analyzed in this study.

However, it did not exceed the critical value for the green PA 6. So, it may explain why the surface of the black material has suffered severe damage after the wear tests, while the green performed well and did not show any significant plastic deformation or damage on its surface. Although there is no data available about the PV limit of the white material, it is likely to be in the same order of magnitude of the black. Thus, the severe damage observed on the surface of this material may also be caused by exceeding the PV limit.

6. Results of the stick-slip tests

Before processing the data of the section III (stick-slip tests), some friction curves were plotted aiming to classify the sliding mode, and so assure that stick-slip motion was in fact occurring. According to Gao et al., 1994, [30], five different sliding modes can occur when two surfaces are rubbed together: (i) smooth sliding, (ii) stick-slip, (iii) near harmonic oscillation, (iv) negative stick-slip, and (v) irregular stick-slip. Fig. 13 shows typical friction-time curves obtained in the section III. A classical stick-slip behavior (saw-tooth pattern) can be seen for the black and green PA 6.

Table 6 presents the data obtained in the stick-slip tests carried out in the present investigation. In this table, the following acronyms were used for representing the output variables: number of stick-slip events in a run of 120s (N_{ev}), time interval corresponding to the slip phase (t_{slip}), mean static coefficient of friction (μ_s), mean kinetic coefficient of friction (μ_k), difference between static and kinetic coefficient of friction ($\Delta\mu$), mean difference between the torque measured in the stick and slip phases (ΔT), slip power (sp), and summation of slip power ($SP_{summation}$).

From Table 6, it can be seen that black PA 6 exhibited occurrences of stick-slip oscillations in all tested conditions. In case of the green plastic, stick-slip motion ceased in contact pressures lower than 1.181 MPa. Besides, both materials showed a clear

Table 5
Data obtained in the PV limit tests for the green and black PA 6.

Green PA 6					Black PA 6				
Normal force [N]	Contact pressure - P [MPa]	Speed rotation [RPM]	Sliding velocity - V [m/s]	PV [MPa m/s]	Normal force [N]	Contact pressure - P [MPa]	Speed rotation [RPM]	Sliding velocity - V [m/s]	PV [MPa m/s]
300	1.181	600	2.51	2.964	200	0.787	250	1.05	0.826
500	1.969	300	1.26	2.481	300	1.181	175	0.73	0.862
600	2.362	275	1.15	2.716	400	1.575	125	0.52	0.819
Mean PV = 2.721					Mean PV = 0.836				
Standard deviation = 0.197					Standard deviation = 0.019				

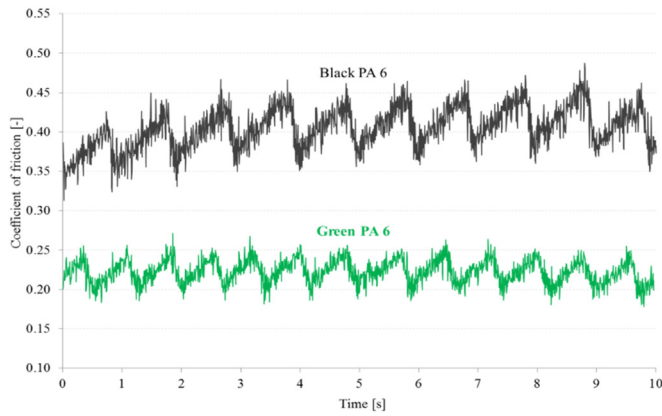


Fig. 13. Friction-time curves showing classical stick-slip motion for the green and black PA 6 analyzed in this study.

reduction in the number of events and slip time as the contact pressure decreased. Data results also showed a trend to reduce $\Delta\mu$, ΔT and sp as the contact pressure decreases.

It is known from literature [31] that the intensity of the stick-slip fluctuation increases with higher differences between static and kinetic coefficient of friction, or $\Delta\mu$. This parameter is mostly used (and well accepted, too) to characterize the amplitude of vibration resulting in brake friction materials, as can be observed in many papers [32–39]. Considering the range of normal force where both plastics exhibited stick-slip motion (from 400 N to 600 N), means $\Delta\mu$ of 0.03 and 0.05 were determined for the green and black plastics, respectively. Based on that result, it can be concluded that the intensity of stick-slip oscillation is lower for green PA 6 than black PA 6.

Recent studies of the current research group [11,12] has presented an alternative parameter for quantifying the stick-slip oscillations resulting in sliding contact surfaces. The alternative parameter is slip power (sp), which is given by dividing the delta torque (ΔT) by the time interval corresponding to the slip period (t_{slip}), as shown in Eq. (2).

$$sp \left[\frac{\text{Nm}}{\text{s}} \right] = \frac{\Delta T}{t_{slip}} \quad (2)$$

As slip power is calculated point-to-point (i.e. a single value is determined for each contact pressure), it was created the summation of slip power (SP summation). This method takes into account all sp values contained in the interval between a null contact pressure ($F_N=0$ N) and the maximum contact pressure, which corresponds to 2.362 MPa ($F_N=600$ N) in the present work. By means of this method, materials that produce stick-slip in a wider

range of operation are penalized.

Eq. (3) presents the SP summation used to quantify the stick-slip oscillations of the plastics selected in this study.

$$SP \text{ summation} = \sum_{F_N=0}^{F_N=600\text{N}} (sp) \quad (3)$$

Applying the SP summation to plastics used in this work, it results 5.37 Nm/s and 7.36 Nm/s for the green and black plastics, respectively. It means that, considering the range of operation analyzed in this study, the black plastic presents 37% more stick-slip oscillations than the green material.

7. Conclusions

The experiments and analysis conducted in the present work allowed to investigate and compare the friction and wear behavior of a bronze alloy and three others PA 6 materials. Using optical microscopy analysis, the most important wear mechanisms of the materials selected in this study were successfully described. This analysis revealed abrasion as the most important wear mechanism for the PA 6 filled with the special solid lubricant. On the other hand, severe damage was observed on the surface of the natural PA 6 and the PA 6 filled with MoS_2 in the wear tests. Bronze alloy, which exhibited the highest wear rate among the materials studied in this work, showed adhesion as the dominant wear mechanism.

Among the plastics selected in this work, PA 6 filled with a special solid lubricant has shown superior tribological properties, since it exhibited a relatively low and stable coefficient of friction, as well as the highest wear resistance and PV limit.

The friction induced oscillations (stick-slip) resulting from low speed tests was also measured in this work. The data were presented in terms of many parameters, such as: number of stick-slip events, difference between static and kinetic coefficient of friction ($\Delta\mu$), mean difference between the torque measured in the stick and slip phases (ΔT), slip power (sp), and summation of slip power (SP summation). Results of both $\Delta\mu$ and SP summation have led to conclude that the green plastic exhibits lower stick-slip oscillations than the black.

Finally, as the PA 6 filled with special lubricant (green plastic) has shown superior tribological properties in all tests carried out in this work, and given that the wear tests were based on the operating conditions of a rotating shaft-bearing system, it can be concluded that this plastic has potential to be used as a bearing bush in industrial applications.

Table 6

Data results obtained in the stick-slip tests.

Material	F_N [N]	Contact pressure MPa	Occurrence of stick-slip motion? [Yes or Not]	N_{ev} [-]	t_{slip} [s]	μ_s [-]	μ_k [-]	$\Delta\mu$ [-]	ΔT [Nm]	sp [Nm/s]	SP summation [Nm/s]
GREEN	600	2.362	Yes	147	0.35	0.26	0.22	0.04	0.73	2.09	5.37
	500	1.696	Yes	156	0.33	0.25	0.22	0.03	0.60	1.82	
	400	1.575	Yes	103	0.32	0.26	0.23	0.03	0.47	1.47	
	300	1.181	Not	–	–	–	0.24	–	–	–	
	200	0.787	Not	–	–	–	0.23	–	–	–	
BLACK	600	2.362	Yes	163	0.54	0.44	0.38	0.06	0.61	1.13	7.36
	500	1.696	Yes	145	0.44	0.40	0.34	0.06	0.76	1.73	
	400	1.575	Yes	130	0.42	0.36	0.32	0.04	0.75	1.79	
	300	1.181	Yes	124	0.42	0.35	0.32	0.03	0.63	1.50	
	200	0.787	Yes	66	0.41	0.37	0.35	0.02	0.50	1.22	

References

- [1] M.D. Bermúdez, F.J. Carrion-Vilches, I. Martinez-Mateo, G. Martinez-Nicolas, Comparative study of the tribological properties of polyamide 6 filled with molybdenum disulfide and liquid crystalline additives, *J. Appl. Polym. Sci.* 81 (2001) 2426–2432.
- [2] F. Song, Q. Wang, T. Wang, The effects of crystallinity on the mechanical properties and the limiting PV (pressure $\times$ velocity) value of PTFE, *Tribol. Int.* 93 (2015) 1–10.
- [3] M.I. Kohan, *Nylon Plastics Handbook*, Hanser, Munich, 1995.
- [4] M. Andó, J. Sukumaran, V. Rodriguez, P. Neis, G. Kalácska, T. Czigány, P. De Baets, Development of new PA6 composites, *Mech. Eng. Lett.: Res. Dev.* 5 (5) (2011) 145–154.
- [5] J.A. Brydson, *Plastics Materials*, 7th edition, 1999.
- [6] R.C.L. Silva, Desgaste de polímeros estruturais de engenharia em contato de deslizamento com cilindro metálico, (Ph.D. thesis) (In Portuguese), Universidade Federal do Rio Grande do Norte, Brazil, 2010.
- [7] S.M. Laranjeira, Estudo do Comportamento ao Atrito e Desgaste de Poliamidas, Master Dissertation (In Portuguese), Universidade de Aveiro, Portugal, 2011.
- [8] F. Van De Velde, P. De Baets, The friction and wear behaviour of polyamide 6 sliding against steel at low velocity under very high contact pressures, *Wear* 209 (1997) 106–114.
- [9] Ensinger, Available on: (<https://www.curbellplastics.com/Research-Solutions/Technical-Resources/Technical-Resources/Engineering-Plastics-Manual-Ensinger>), (accessed January 2017).
- [10] D. Masotti, N.F. Ferreira, P.D. Neis, J. Poletto, A.R. Menetrier, G. Tonet, L.T. Matoso, P.E. Varante, R.J. Canali, Experimental evaluation of creep groan noise using a laboratory-scaled tribometer, Eurobrake 2014, paper number: EB2014-ST-005, Dresden, Germany, 2014.
- [11] P.D. Neis, N.F. Ferreira, L.T. Matoso, D. Masotti, J.C. Poletto, Definition of a suitable parameter for characterizing creep groan propensity in brake pads, SAE Technical paper (ISSN 0148-7191), presented in 12th SAE Brazil International Brake Colloquium & Engineering Display, Brazil, 2015.
- [12] P.D. Neis, N.F. Ferreira, J.C. Poletto, L.T. Matoso, D. Masotti, Quantification of brake creep groan in vehicle tests and its relation with stick-slip obtained in laboratory tests (ISSN: 0022-460X), *J. Sound Vib.* 369 (2016) 63–76.
- [13] P.D. Neis, Development of a multifunctional tribometer with independent temperature control of the brake disk, (Ph.D. thesis), Joint PhD between Federal University of Rio Grande do Sul (Brazil) and Ghent University (Belgium), 2012.
- [14] P.D. Neis, Projeto e construção de um tribômetro para estudos relacionados a materiais de fricção veiculares, Master Dissertation (In Portuguese), Federal University of Rio Grande do Sul, Brazil, (<http://hdl.handle.net/10183/107628>), 2008.
- [15] P.D. Neis, N.F. Ferreira, A multifunctional tribometer for friction characterization, SAE Technical Paper 2013-36-0035, <http://dx.doi.org/10.4271/2013-36-0035>, 2013.
- [16] P.D. Neis, N.F. Ferreira, F.J. Lorini, Contribution to perform high temperature tests (fading) on a laboratory-scale tribometer, *Wear* 271 (2011) 2660–2664.
- [17] R.G. Bayer, *Engineering Design for Wear*, 2nd edition, CRC Press, 2004.
- [18] F. Van, de Velde, P. De Baets, The relation between friction force and relative speed during the slip-phase of a stick-slip cycle, *Wear* 219 (1998) 220–226 (1998).
- [19] W. Ost, P. De Baets, M. Loccufier, S.V. Autrève, Torsional vibrations of wet clutches: influence of mechanical parameters, in *Meccanica Journal*, Italy, 2009.
- [20] W. Ost, *Tribology and Vibrations of Wet Clutches* (PhD Thesis), Ghent University, Belgium, 2009.
- [21] P.D. Neis, P. De Baets, W. Ost, Y. Perez Delgado, M. Loccufier, F. Al-Bender, N. F. Ferreira, F.J. Lorini, Investigation of the dynamic response in a dry friction process using a rotating stick-slip tester, *Wear* 271 (2011) 2640–2650.
- [22] C.W. Park, M.W. Shin, H. Jang, Friction-induced stick-slip intensified by corrosion of gray iron brake disc, *Wear* 309 (2014) 89–95.
- [23] R. Rabinowicz, *Friction and Wear of Materials*, second ed, Wiley, New York, 1995.
- [24] W. Sextro, *Dynamical Contact Problems with Friction: Models, Methods, Experiments and Application*, 2nd ed, Springer, Berlin, 2007.
- [25] P.J. Blau, B.C. Jolly, Wear of truck brake lining materials using three different test methods, *Wear* 259 (2005) 1022–1030.
- [26] L.M.S.C. Seabra, Comportamento tribológico de polímeros alimentares contra aço inoxidável em escorregamento não lubrificado, in portuguese, Master dissertation (In Portuguese), Faculdade de Engenharia da Universidade do Porto, Portugal, 2000.
- [27] M. Fellah, M. Labaiz, O. Assala, L. Dekhil, A. Taleb, H. Rezag, A. Iost, Tribological behavior of Ti-6Al-4V and Ti-6Al-7Nb alloys for total hip prosthesis, *Adv. Tribol. Volume* 2014 (2014) 1–13, <http://dx.doi.org/10.1155/2014/451387>.
- [28] J.C. Gerdeen, R.A.L. Rorrer, *Engineering Design with Polymers and Composites*, 2nd edition, CRC/Taylor & Francis, 2012.
- [29] K.C. Ludema, *Friction, Wear, Lubrication*, CRC Press, 1996.
- [30] C. Gao, D. Kuhlmann-Wilsdorf, D.D. Makel, The dynamic analysis of stick-slip motion, *Wear* 173 (1994) 1–12.
- [31] F.P. Bowden, D. Tabor, *The Friction and Lubrication of Solids*, Oxford University Press, Oxford, 1953.
- [32] S.W. Yoon, M.W. Shin, W.G. Lee, H. Jang, Effect of surface contact conditions on the stick-slip behavior of brake friction material, *Wear* 294–295 (2012) 305–312.
- [33] Z. Fuadi, S. Maegawa, K. Nakano, K. Adachi, Map of low-frequency stick-slip of a creep groan, *Proc. Inst. Mech. Eng. Part 1: J. Eng. Tribol.* 224 (2010) 1235–1246.
- [34] H. Jang, J.S. Lee, J.W. Fash, Compositional effects of the brake friction material on creep groan phenomena, *Wear* 251 (2001) 1477–1483.
- [35] C.W. Park, M.W. Shin, H. Jang, Friction-induced stick-slip intensified by corrosion of gray iron brake disc, *Wear* 309 (2014) 89–95.
- [36] Z. Fuadi, K. Adachi, H. Ikeda, H. Naito, K. Kato, Effect of contact stiffness on creep-groan occurrence on a simple caliper-slider experimental model, *Tribol. Lett.* 33 (2009) 169–178, <http://dx.doi.org/10.1007/s11249-008-9404-4>.
- [37] W.K. Lee, T.H. Rhee, H.S. Kim, H. Jang, Effects of antimony trisulfide (Sb₂S₃) on sliding friction of automotive brake friction materials, *Met. Mater. Int.* 19 (n. 5) (2013) 1101–1107, <http://dx.doi.org/10.1007/s12540-013-5027-x>.
- [38] S.M. Lee, M.W. Shin, W.K. Lee, H. Jang, The correlation between contact stiffness and stick-slip of brake friction materials, *Wear* 302 (2013) 1414–1420.
- [39] W.K. Lee, M.W. Shin, S.H. Kim, H. Jang, M.H. Cho, The influence of humidity on the sliding friction of brake friction material, *Wear* 302 (2013) 1397–1403.